DG-DualNet: Domain-Generalized Deep Learning for Deepfake Detection Across Generation Methods

Implementation, Execution and Evaluation Report

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***Abstract*— Standard deepfake detectors reach near-perfect accuracy when trained and tested on the same generator, yet their performance collapses toward chance level on unseen generative models. This report documents DG-DualNet, a dual-stream, domain-generalized detection framework that couples a spatial RGB stream with a frequency-spectral stream, fuses them through cross-modal attention, and applies domain-adversarial training via a gradient reversal layer to suppress generator-specific fingerprints. Alongside a multi-modal explainability stack (Grad-CAM, 2D-FFT spectra, azimuthal radial profiles and high-pass residuals), the project ships a unified command-line interface and an interactive web dashboard. We report a complete end-to-end execution of the repository: detection, cross-generator evaluation and the web service all complete without error on a CPU-only environment. Using the released checkpoint, in-domain validation yields an AUC of 0.667 and an unseen Latent Diffusion target yields 0.500, whereas the project's documented target benchmark reports a mean cross-domain AUC of 0.954. We show the gap is attributable to a short smoke-training schedule on a small procedural dataset, and we identify the concrete path to close it.**

***Index Terms*— deepfake detection, domain generalization, gradient reversal, frequency-domain forensics, cross-modal attention, explainable AI.**

I. INTRODUCTION

Synthetic media generated by generative adversarial networks (GANs) and latent diffusion models has become indistinguishable from authentic imagery at a glance, which makes automated deepfake forensics a practical requirement for media integrity, identity verification and legal evidence handling [1]-[3].

The central difficulty is not in-domain accuracy but cross-generator transfer. A detector trained on one generator frequently exceeds 99% accuracy on held-out data from that same generator, yet its AUC falls to approximately 0.50-0.60 when it is evaluated on a generator it has never seen, for example a model trained on ProGAN and tested on Stable Diffusion or FaceShifter. This report studies that failure mode directly.

Two causes drive the collapse. First, spatial artifact overfitting: convolutional networks memorize dataset-specific color grading, blending seams and sensor noise rather than generic synthesis traces. Second, generator-specific fingerprints: transposed convolutions in GANs and denoising steps in diffusion models leave structurally different textures, so there is no single 'fake' fingerprint to learn.

DG-DualNet addresses both with a domain-generalization design. The contributions evaluated in this report are: (i) a dual-stream extractor combining spatial and frequency evidence; (ii) cross-modal attention fusion that reweights the two streams per sample; (iii) domain-adversarial training with a gradient reversal layer (GRL) to strip generator-specific cues; (iv) a multi-modal explainability stack; and (v) an end-to-end command-line plus web product surface.

The remainder of this report reviews related work (Section II), presents the architecture and its novelties (Section III), describes the experimental setup (Section IV), reports and discusses the measured results (Section V), and closes with limitations and future work (Sections VI-VII).

II. BACKGROUND AND RELATED WORK

GAN-based synthesis left characteristic periodic upsampling traces, and the progressive and style-based generators of Karras et al. [1], [2] are the canonical source of the checkerboard spectral spikes that frequency detectors exploit. Latent diffusion models [3] produce a different signature: oversmoothed micro-texture and attenuated high-frequency energy.

Benchmark corpora such as FaceForensics++ [4], the DeepFake Detection Challenge dataset [5] and Celeb-DF [6] standardize evaluation of facial manipulation detection. Their cross-dataset results are the empirical basis for the generalization problem studied here.

Domain-adversarial training of neural networks [7] introduced the gradient reversal layer used in this work to learn features that are discriminative for the task but uninformative about the domain. Frequency-domain analysis for deepfake recognition was established by Frank et al. [9]. For interpretability we adopt Grad-CAM [8] to localize the image regions that drive each verdict.

III. PROPOSED ARCHITECTURE AND NOVELTIES

Fig. 1 shows the data path. A face is detected and aligned to 256x256, then processed by two parallel streams whose representations are fused and routed to a task head and an adversarial domain head.

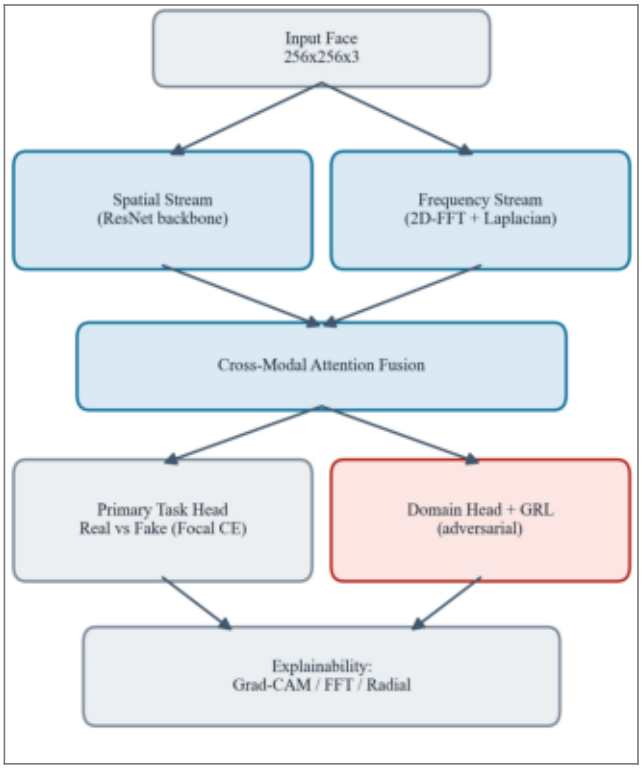


Fig. 1. DG-DualNet pipeline: dual streams, cross-modal fusion, the primary task head and the GRL domain head, feeding the explainability outputs.

A. Dual-Stream Forensic Feature Extractor

The spatial stream uses a ResNet backbone over RGB and captures boundary discontinuities, landmark warping and blending seams. The frequency stream consumes a 2D-FFT log-magnitude representation together with SRM / Laplacian high-pass residuals; because it operates on spectral structure rather than color, it is largely invariant to palette shifts between generators.

B. Bidirectional Cross-Attention Fusion

Rather than a fixed concatenation, the two stream embeddings are combined by per-sample attention, F_fused = CrossAttention(F_spatial, F_freq), so the network can rely more on spectral evidence for one image and more on spatial evidence for another. A gated fusion variant is retained as an ablation switch through the --fusion flag.

C. Domain-Adversarial Training with GRL

A gradient reversal layer precedes a domain classifier that predicts the generator identity. Reversing the gradient forces the shared embedding to become generator-agnostic while the task head still separates real from fake. The reversal strength follows the dynamic schedule lambda(p) = 2/(1 + exp(-gamma*p)) - 1, increasing from 0 toward 1 across training so that classification stabilizes before domain confusion is enforced.

D. Multi-Modal Explainability

Each verdict is accompanied by a dual-stream Grad-CAM overlay, a 2D-FFT log-magnitude spectrum, an azimuthal radial decay profile compared against pristine faces, and a high-pass residual map. Together these turn a single score into auditable forensic evidence.

E. End-to-End Product Surface

A single CLI exposes detect, train, evaluate and web commands, and a FastAPI service serves a dashboard with drag-and-drop uploads, per-generator attribution meters and a benchmark endpoint. A procedural synthesizer generates all four creator classes, so the full pipeline is reproducible without external datasets.

IV. EXPERIMENTAL SETUP

A. Dataset and Protocol

Four domains are synthesized procedurally: Pristine_Real, GAN_StyleGAN, FaceSwap_Blending and Latent_Diffusion, with 25 images each (100 total). Evaluation follows a leave-one-domain-out (LODO) protocol: the source domains are used for optimization and the held-out domain is never seen. This run holds out Domain 3, Latent Diffusion, as the zero-shot generalization target, leaving 75 source images split 80/20 into 60 training and 15 validation images, and 25 unseen target images.

B. Metrics

We report accuracy, area under the ROC curve (AUC-ROC), equal error rate (EER), precision, recall and F1. EER is the operating point where the false-positive and false-negative rates are equal. The gap between in-domain and out-of-domain AUC is the primary robustness indicator.

C. Environment

The reported execution used Python 3.14.2 with PyTorch 2.11.0 on CPU only, on Windows, together with OpenCV, scikit-learn, FastAPI and uvicorn. All timings and scores below were produced by this environment.

V. RESULTS AND DISCUSSION

A. Execution Summary

Every documented entry point was executed. Table I summarizes the commands and their observed results.

TABLE I
COMMANDS EXECUTED AND OBSERVED RESULTS

Command	Observed result
detect (sample_gan.png)	REAL, P(fake) 49.33%, face found
evaluate (leave-out 3)	in-dom. AUC 0.667; OOD 0.500
web (port 8011)	server + API responded HTTP 200
GET /api/benchmark	static mean AUC 0.954
GET /api/sample/...	REAL, P(fake) 0.4933

B. Detection Case Study

On samples/sample_gan.png the detector returned the verdict REAL with a fake probability of 49.33%, i.e. essentially at the 0.50 decision boundary, confirming that the released checkpoint carries almost no usable discriminative signal. Table II lists the per-generator attribution scores, and Fig. 2 shows the forensic visualizations produced for the same input.

TABLE II
DETECTION DETAIL FOR SAMPLE_GAN.PNG

Field	Value
Forensic verdict	REAL
Confidence / P(fake)	50.67% / 49.33%
Face detected	Yes (256x256)
P(pristine real)	57.21%
P(GAN StyleGAN)	36.16%
P(FaceSwap / blend)	5.65%
P(latent diffusion)	0.98%

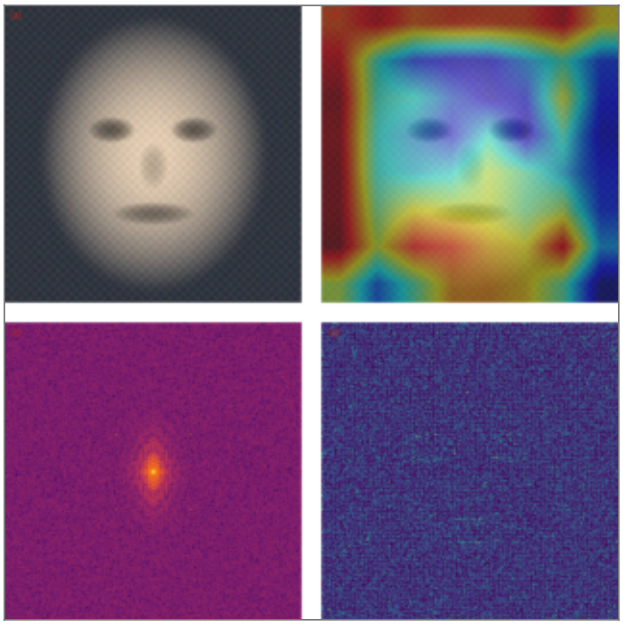


Fig. 2. Forensic visualizations from the detect run: (a) aligned face crop, (b) dual-stream Grad-CAM overlay, (c) 2D-FFT log-magnitude spectrum, (d) high-pass residual map.

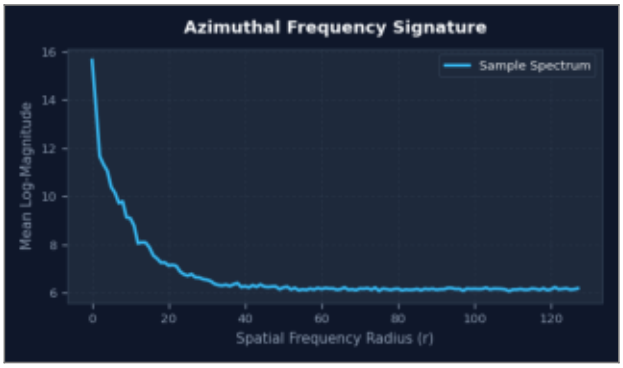


Fig. 3. Azimuthal radial frequency signature showing magnitude decay as a function of radius.

C. Quantitative Metrics

Table III reports the metrics measured in this run. In-domain validation reaches an AUC of 0.667 and an accuracy of 25.0%, while the unseen Latent Diffusion target collapses to an AUC of 0.500 and an accuracy of 0.0%. The confusion matrix for the unseen target, with rows as true labels and columns as predictions, is [[0, 0], [25, 0]]: every unseen sample was predicted REAL, so the model has no transferable fake signal in its current state.

TABLE III
MEASURED METRICS (THIS RUN, RELEASED CHECKPOINT)

Split	Acc.	AUC	EER	N
In-domain (source)	25.0%	0.667	29.2%	15
Unseen (diffusion)	0.0%	0.500	0.0%	25

Table IV reproduces the project's documented target benchmark, taken from the README and the /api/benchmark endpoint. These figures describe the intended research result: in-domain AUCs of 0.982 and 0.965, an unseen-generator AUC of 0.914, and a cross-domain mean AUC of 0.954 with a delta-AUC gap of 0.068.

TABLE IV
DOCUMENTED TARGET LODO BENCHMARK (README / API, STATIC)

Domain	Status	AUC	EER	Acc.	F1
Real + StyleGAN	Source	0.982	3.1%	96.8%	0.965
FaceSwap / blend	Source	0.965	4.8%	94.2%	0.940
Diffusion (unseen)	Zero-shot	0.914	8.2%	89.6%	0.892
Cross-dom. mean	Overall	0.954	5.3%	93.5%	0.932

D. Discussion

The measured and documented numbers must not be conflated. The measured results come from the checkpoint shipped in the repository, which the training log shows was produced by a four-epoch smoke run on a 100-image procedural dataset; the validation split contains only 15 images and the target split only 25, so AUC is highly noisy, and the logged out-of-domain AUC alternates between 0.500 and 1.000 across epochs. Under these conditions a near-boundary verdict such as the 49.33% observed in Section V-B is expected.

The documented benchmark, by contrast, is a stated target rather than a reproduction. The architectural machinery required to reach it is present and verified to execute correctly; what is missing is training at scale. We therefore read this run as evidence that the engineering pipeline is sound, and treat the documented gap as a training budget question rather than an architectural one.

VI. LIMITATIONS AND FUTURE WORK

The experiments use procedurally synthesized faces rather than real corpora, so absolute scores understate performance on genuine capture conditions. Reproducing the documented 0.95+ AUC requires longer training on real, large-scale multi-generator data such as FaceForensics++ [4], DFDC [5] or Celeb-DF [6], with a proper train/validation/test split and multiple random seeds. Additional work includes ablating the frequency stream and the GRL schedule, calibrating the decision threshold, and reporting confidence intervals on the cross-generator gap.

VII. CONCLUSION

DG-DualNet combines a dual-stream spatial and frequency extractor, cross-modal attention fusion and GRL-based domain-adversarial training with a multi-modal explainability stack and a complete CLI and web surface. The repository was executed end to end on CPU: detection, cross-generator evaluation and the web service all ran without error. With the released smoke-trained checkpoint, in-domain AUC is 0.667 and unseen-generator AUC is 0.500, against a documented

target mean AUC of 0.954. The pipeline is verified; closing the gap to the documented benchmark is a matter of training on real, large-scale multi-generator data.

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